MILITARY SPECIFICATION

CLOTH, FELT-FABRIC COMPOSITE, UNDERCOLLAR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements for one type of felt-fabric composite undercollar cloth.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-1132	- Packaging and Packing of Woolen, Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics

MILITARY

MIL-C-43665	- Cloth, Wool; Mothproofing Treatment of

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:39Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE WOOL PRODUCTS LABELING ACT OF 1939

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Standard sample The finished cloth shall match the standard sample for shade, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

*3.2 First article When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.3 Materials (see 6.4)

3.3.1 Base cloth (scrim) The yarn for the base cloth shall be cotton which has been cleaned, carded, and spun into singles for both the warp and filling. Testing shall be performed as specified in 4.5.

3.3.1.1 Physical requirements The base cloth shall conform to the requirements listed in Table I when tested as specified in 4.5.

Table I - Base cloth physical requirements

Weight oz/sq yd (g/m^2)		Yarns per inch (cm) (Min.)		Thickness (inch) (cm)		
(Min)	(Max.)	Warp	Filling	(Min.)	(Max.)	Weave
4.0 (136)	4.4 (149)	42 (16.5)	26 (10.2)	.016 (.041)	.020 (.051)	Plain

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:39Z
Check the source to verify that this is the current version before use.

*3.3.1.2 Alternate scrim As an alternate to the cotton base scrim specified in 3.3.1 and 3.3.1.1, a rayon scrim may be used. When this option is exercised, the weight shall be 3.5 to 3.9 oz/sq yd (119 to 132 g/m^2) and the yarns/inch (min.) shall be 40 (15.7 yarns/cm) for the warp and 24 (9.4 yarns/cm) for the filling.

3.3.2 Felt-fabric composite The felt-fabric composite shall be comprised of the following (see 3.3.2.1 thru 3.3.2.4).

* 3.3.2.1 Wool The wool shall be fleece or pulled wool or a combination thereof. The use of reprocessed wool, reused wool and noils is permitted.

3.3.2.2 Felt-web The fiber for the felt-web shall be not less than 40 percent wool, and the remainder shall be rayon and/or cotton (based on the dry weight).

3.3.2.3 Felt-fabric The scrim (see 3.3.1 and 3.3.1.2) shall be laid between two layers of felt-web (see 3.3.2.2) and needled together to form a felt-fabric composite.

*3.3.2.4 Physical requirements The felt-fabric composite shall conform to the requirements in Table II when tested as specified in 4.5.

Table II - Felt-fabric composite physical requirements

Characteristics	Requirements	
	Min.	Max.
Weight, oz/sq yd (g/m^2)	9.0 (305)	12.0 (407)
Breaking strength, lbs (Newton)		
Machine direction	50 (223)	-
Cross direction	30 (134)	-
Thickness, inches (cm)	.040 (.102)	.060 (.152)
Stiffness, inch (cm) (drape)		
Machine direction	1.25 (3.18)	2.5 (6.35)
Cross direction	1.25 (3.18)	2.5 (6.35)
Shrinkage in sponging		
Machine direction	-	5.0%
Cross direction	-	5.0%
Shrinkage in wet-dry cleaning		
Machine	-	2.0%
Cross direction	-	2.0%

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:39Z
Check the source to verify that this is the current version before use.

3.4 Color The color of the cloth shall be as specified (see 6.2) and shall match the standard sample.

*3.4.1 Matching The color of the finished cloth shall match the standard sample under artificial daylight having a temperature of 7500 Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800 Kelvin.

3.4.2 Colorfastness The finished cloth shall show fastness to wet-dry cleaning, perspiration, and crocking equal to or better than the standard sample when tested as specified in 4.5. When no standard sample is available, the cloth shall show a minimum of "good" fastness to wet-dry cleaning and perspiration and shall show a Munsell Value for crocking not lower than 8.5.

3.5 Width The width of the cloth shall be as specified (see 6.2).

3.6 Finish The needled felt-fabric composite shall be felted, scoured, piece dyed, and compressed and shall have a finish equal to that shown by the standard sample.

3.6.1 pH The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.0 when tested as specified in 4.5.

3.7 Length and put-up Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards (36.6m). The cloth shall be put up in full width rolls with face side inward as specified in PPP-P-1132.

3.8 Wool content label Cloth manufactured under this specification shall be labeled in accordance with the Wool Products Labeling Act of 1939.

3.9 Mothproofing The cloth shall be given a mothproofing treatment in accordance with MIL-C-43665.

3.10 Marking The back of the cloth shall be marked "Back" at both ends of each roll with letters not less than 1/2 inch (1.3cm) in height. The marking shall be clearly legible in any indelible marking medium used commercially.

3.11 Workmanship The finished cloth shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:39Z
Check the source to verify that this is the current version before use.

4.1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

*4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection When required, the first article of finished cloth shall be visually inspected for color, finish and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in the specification or applicable purchase document.

4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.3.

4.4.2.1 Yard-by-yard examination The required yardage of each roll shall be examined on the face side only, and defects clearly noticeable at normal inspection distance (3 feet) (.9m) shall be classified as listed. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the most serious defect shall be counted. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The sample unit shall be one linear yard (.9m). The acceptable quality level shall be 4.0 defects per 100 units for all defects listed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:39Z
Check the source to verify that this is the current version before use.

The sample size shall be in accordance with inspection level II of MIL-STD-105. The lot size shall be expressed in units of one yard (.9m) each. The number of rolls selected shall be in accordance with Table III. An approximately equal number of yards shall be examined from each roll in the sample.

Defects

Hole, cut, tear or break
Abrasion or rub mark resulting in damage to fibers
Crease or wrinkle, hard embedded
Lump
Spot, stain or streak of a permanent nature
Embedded foreign matter
Surface pucker or ripple

4.4.2.2 Overall examination Each defect shall be counted no more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in Table III.

Defects

Width less than specified 1/
Overall uncleanness 1/
Objectionable odor
Off shade, shaded side to side, side to center or end to end
Degree and character of finish not equal to standard sample
Wrong color, i.e., not as specified

1/ It shall not be scored unless it occurs for more than 5 yards (4.6m) per roll.

Table III - Sample size

Lot size (yards)	Sample size (rolls)	Acceptance number
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:39Z
Check the source to verify that this is the current version before use.

4.4.2.3 Examination for length

4.4.2.3.1 Individual rolls The rolls shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards (1.8m) less than the length marked on the ticket shall be considered a defect with respect to length. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the acceptance number shall be as shown in Table III.

4.4.2.3.2 Total yardage in sample The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.4.2.4 Examination for back marking, back direction on roll, and compliance with the Wool Products Labeling Act During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Back stamping missing from either or both ends.
Back stamping on wrong side.
Not rolled with face side inward.
Letters less than 1/2 inch (1.3cm) in height.
Not labeled or ticketed in accordance with the Rules and Regulations Under the Wool Products Labeling Act.

4.4.3 Examination of Packaging requirements An examination shall be made in accordance with the provisions of PPP-P-1132 to determine that packaging, packing and marking comply with the Section 5 requirements.

4.5 Testing of the end item The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table IV shall be followed. The physical and chemical values specified in Section 3, except when otherwise indicated, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The sample unit of the base (scrim) cloth shall be 1/2 yard (.45m) full width. The sample unit shall be 1 yard (.9m) full width of the finished felt-fabric composite, and the lot shall be expressed in units of 1 yard (.9m). The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:39Z
Check the source to verify that this is the current version before use.

Table IV - Test methods

Characteristic	Requirement paragraph	Test Method
Base Cloth (Scrim)		
Cotton identification	3.3.1	1200 1/
Weight, oz/sq yd (g/m^2)	3.3.1.1	5041
Yarns per inch (cm)	3.3.1.1	5050
Thickness (inch) (cm)	3.3.1.1	5030
Weave	3.3.1.1	Visual
Rayon identification	3.3.1.2	1500 1/ 1510 1/ and 1520 1/
Wool Fiber Content (Web)	3.3.2.2	2101 1/
Felt-fabric Composite		
Fiber identification:		
Wool	3.3.2.1	1100 1/
Rayon	3.3.2.2	1500 1/, 1510 1/ and 1520 1/
Cotton	3.3.2.2	1200 1/
Weight, oz/sq yd (g/m^2)	3.3.2.4	5041 2/
Breaking strength (lbs)(N):		
Machine direction	3.3.2.4	5100
Cross direction	3.3.2.4	5100
Thickness (inches) (cm)	3.3.2.4	5030
Stiffness (max) (Drape):		
Machine direction	3.3.2.4	5206
Cross direction	3.3.2.4	5206
Shrinkage in sponging:		
Machine direction	3.3.2.4	5590
Cross direction	3.3.2.4	5590
Shrinkage in wet-dry cleaning:		
Machine direction	3.3.2.4	5580
Cross direction	3.3.2.4	5580
Colorfastness to:		
Wet-dry cleaning	3.4.2	5622
Crocking	3.4.2	5651
Perspiration	3.4.2	5680
pH	3.6.1	2811
Mothproofing	3.9	3/

1/ A certificate of compliance will be acceptable for this characteristic.
2/ Based on overall weight of composite material.
3/ As specified in MIL-C-43665.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:39Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation-packaging Preservation packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1132.

5.2 Packing Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132.

5.3 Marking In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1132.

6. NOTES

6.1 Intended use The cloth is intended for use as a combination interlining/facing for the underside of the collar of uniform coats and overcoats.

6.2 Ordering data Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When first article is required (see 3.2).
(c) Color of cloth required (see 3.4).
(d) Width of cloth required (see 3.5).
(e) Length if other than specified (see 3.7).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

*6.5 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing activity:
Navy - NU

Project No. 8305-0645

Review activities:
Army - MD
Navy - MC, CG
Air Force - 99, 45
DLA - CT

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-29137A CLOTH, FELT-FABRIC COMPOSITE, UNDERCOLLAR

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426 Replaces edition of 1 Jan 72 which may be used. S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:39Z
Check the source to verify that this is the current version before use.

FOLD

OFFICER IN CHARGE
NAVY CLOTHING & TEXTILE RESEARCH FACILITY
21 STRATHMORE ROAD
NATICK, MA 01760

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE NAVY
DoD-316



OFFICER IN CHARGE
NAVY CLOTHING & TEXTILE RESEARCH FACILITY
21 STRATHMORE ROAD
NATICK, MA 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:39Z
Check the source to verify that this is the current version before use.